FSDP2 Execution Architecture for Teacher Scoring

Pinned verl worker ·single GPU ·contract-preserving readout

Request

`input_ids` / attention mask

Student candidate IDs $[\text{batch}, T, K]$

TeacherFSDPWorker

prepare_model_inputs / layout alignment

Validate candidates, mask and offsets

Execution policy

Chunk rows B / readout budget H

Route selection / budget admission

One Qwen2 / FSDP2 outer forward

Tied LM-head weights ·module hooks ·optional CPU offload

Transformer Backbone + Final RMSNorm

Hidden states $\mathbf{H} \in \mathbb{R}^{N \times d}$

One execution path per request

A0 Native Dense

LM head: $\mathbf{HW}^{\mathsf{T}}$

Materialize full logits $[N, V]$

Full logits leave FSDP forward

A1 Memory-Bounded

Per tile $\mathbf{H}_i[B, d] \rightarrow$ LMHead $\rightarrow \mathbf{Z}_i[B, V]$

FP32 LSE / candidate gather / teacher Top-K per tile

Write compact $[N, K]$ buffers; discard $\mathbf{Z}_i$; repeat

$\lceil N/B \rceil$ LM-head calls, bounded live tiles

Reduce after FSDP forward

Full-vocabulary LSE / gather / Top-K

Reduction already completed per tile

Only compact buffers returned; no full logits concatenation

Identical Worker output contract

Candidate logprobs $[N, K]$; teacher Top-K IDs $[N, K]$

CPU nested tensors with valid-row offsets restored

Separate test: pinned reverse-KL loss → Student gradients → restricted RMSNorm step

N: projected rows; V: vocabulary; B: rows per tile. Budget is a readout estimate, not a device cap.